



Department of Textile and Fibre Technology
Indian Institute of Technology, Delhi

Major Project Part I (TXD 401)

Mid-Infrared Imaging and Machine Learning Approaches for Non-Destructive and Efficient Material Detection of Textile Fabrics

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Principal Investigator : Prof. Apurba Das

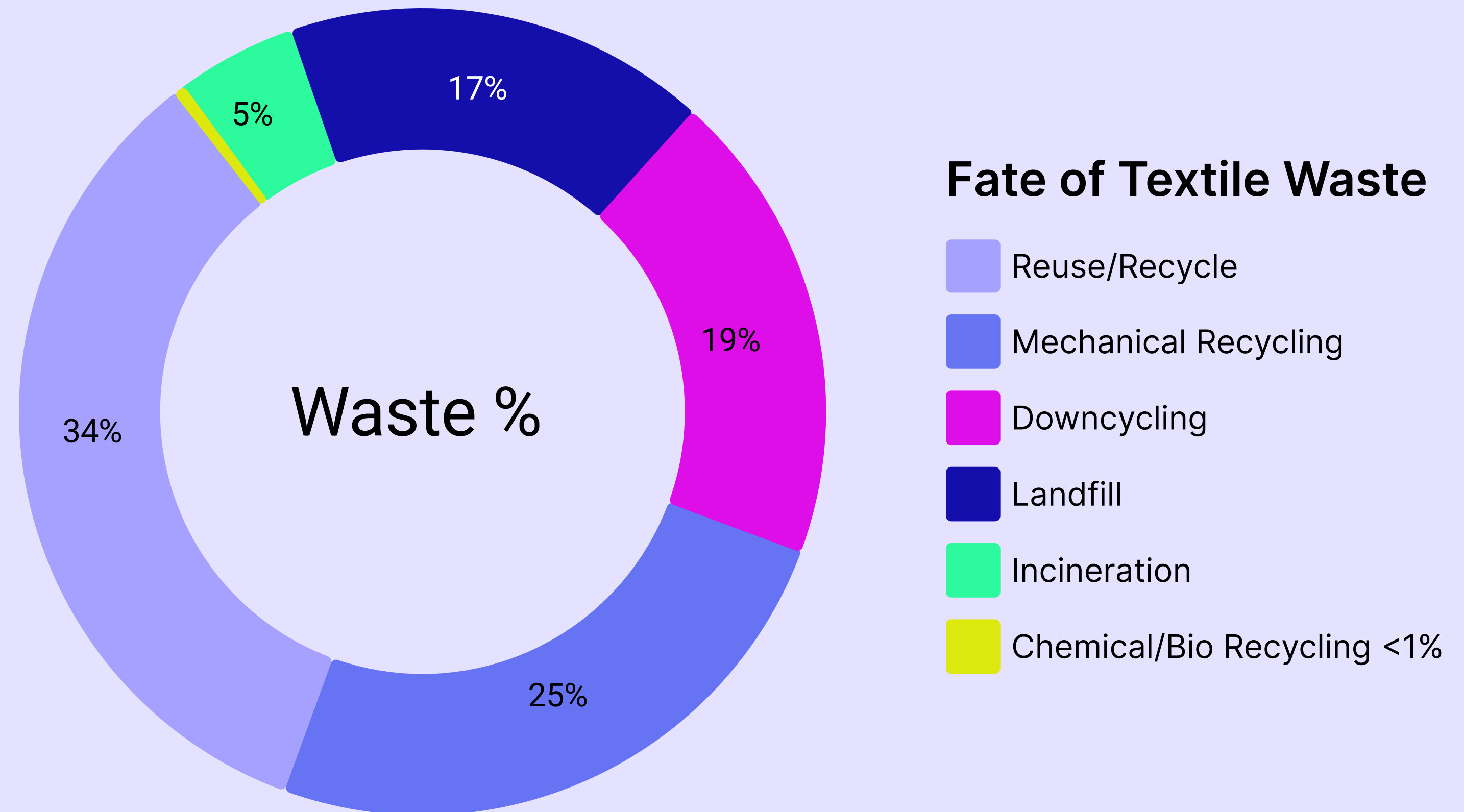
Co Principal Investigator : Prof. Bipin Kumar





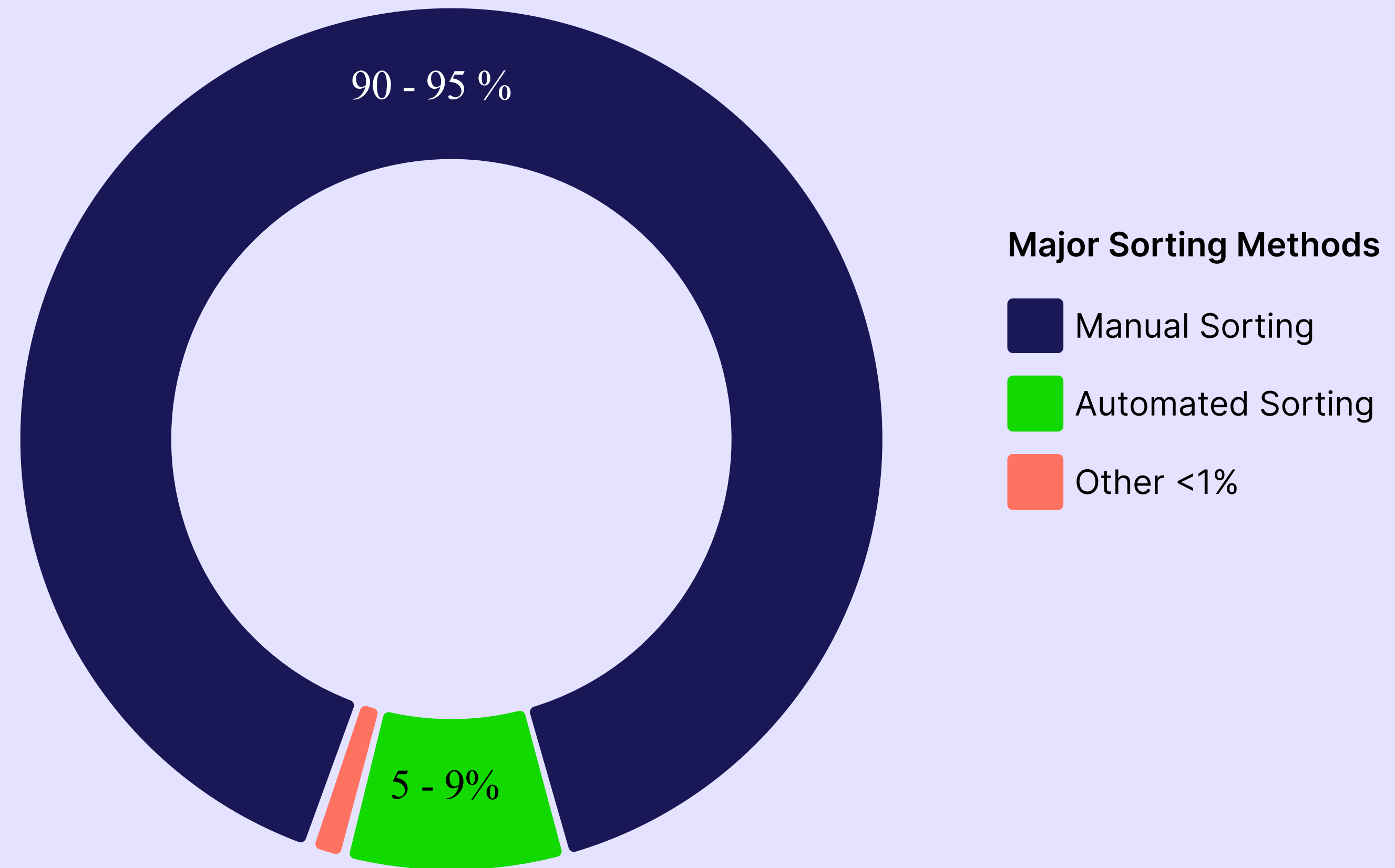
“Do you know that in India, 7.8 million tons of textile waste is generated annually”

What happens with those textile waste in India?



Source : <https://reports.fashionforgood.com/report/sorting-for-circularity-india-wealth-in-waste/chapterdetail?reportid=813&chapter=3>

Distribution of Major Sorting Methods in India



Source : <https://reports.fashionforgood.com/report/sorting-for-circularity-india-wealth-in-waste/chapterdetail?reportid=813&chapter=3>

Literature Review

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Figure 1. A review of the literature on near-infrared (NIR) spectroscopy. The figure is a screenshot of a journal article page. It includes a portrait of Marena Manley, a biographical note, and several paragraphs of text. Two specific text blocks are highlighted with blue and yellow backgrounds, and arrows point from them to larger versions of the same text on the right side of the slide.

Marena Manley was born in 1966 and grew up in the Northern Cape, South Africa. She obtained her BSc in Food Science degree at Stellenbosch University and subsequently her Honours and MSc degrees at the University of Pretoria. She received her PhD (in near-infrared spectroscopy) in 1995 from the University of Plymouth, UK. After 18 months in industry, she joined Stellenbosch University in 1997 as a lecturer where she is currently appointed as Professor. Her research interests involve the application of near-infrared (NIR) spectroscopy, NIR hyperspectral imaging and lately also X-ray micro-computed tomography to study whole grain endosperm texture and other cereal defects.

Fundamental principles of near-infrared spectroscopy

Spectra in the NIR region result from energy absorption by organic molecules, and comprise overtones and combinations of overtones originating from fundamental bond vibrations (stretching or bending) occurring in the mid-infrared (MIR) region of the spectrum. The features in NIR spectra of organic compounds are thus orders of magnitude weaker than those in the MIR. Overtones can be found by dividing the wavelengths in the infrared region by approximately 2, 3 or 4 and provide the advantage of a dilution series. NIR spectra can thus be collected directly on samples without the need for dilution, enabling direct analysis of solid samples. Ease of sample preparation and presentation results in NIR spectroscopy often being used in favour of MIR spectroscopy.

The NIR region extends from 800 to 2500 nm (12 500 to 4000 cm⁻¹; 120 to 375 THz), between the visible from 380 to 780 nm (26 316 to 12 820 cm⁻¹; 385–790 THz) and MIR from 2500 to 15 000 nm (4000 to 400 cm⁻¹; 30 to 120 THz) regions. NIR spectra contain information about the major X–H chemical bonds, i.e. C–H, O–H and N–H. All molecules containing hydrogen will have a measurable NIR spectrum, resulting in a large range of organic materials to be suitable for NIR analysis.

Due to the overtone and combination modes and large numbers of possible vibrations, NIR spectra are very complex, consisting of many overlapping peaks (referred to as ‘multicollinearity’), which result in broad bands. This makes it difficult to interpret NIR spectra visually, assign specific features to specific chemical components or extract information contained in the spectra easily. It was, however, realised early on that, with the use of appropriate regression techniques, relationships between absorption values at specific wavelengths and reference values of the constituent to be predicted could be established. Specific chemical

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Chem. Soc. Rev., 2014, 43, 8200–8214 | 8201

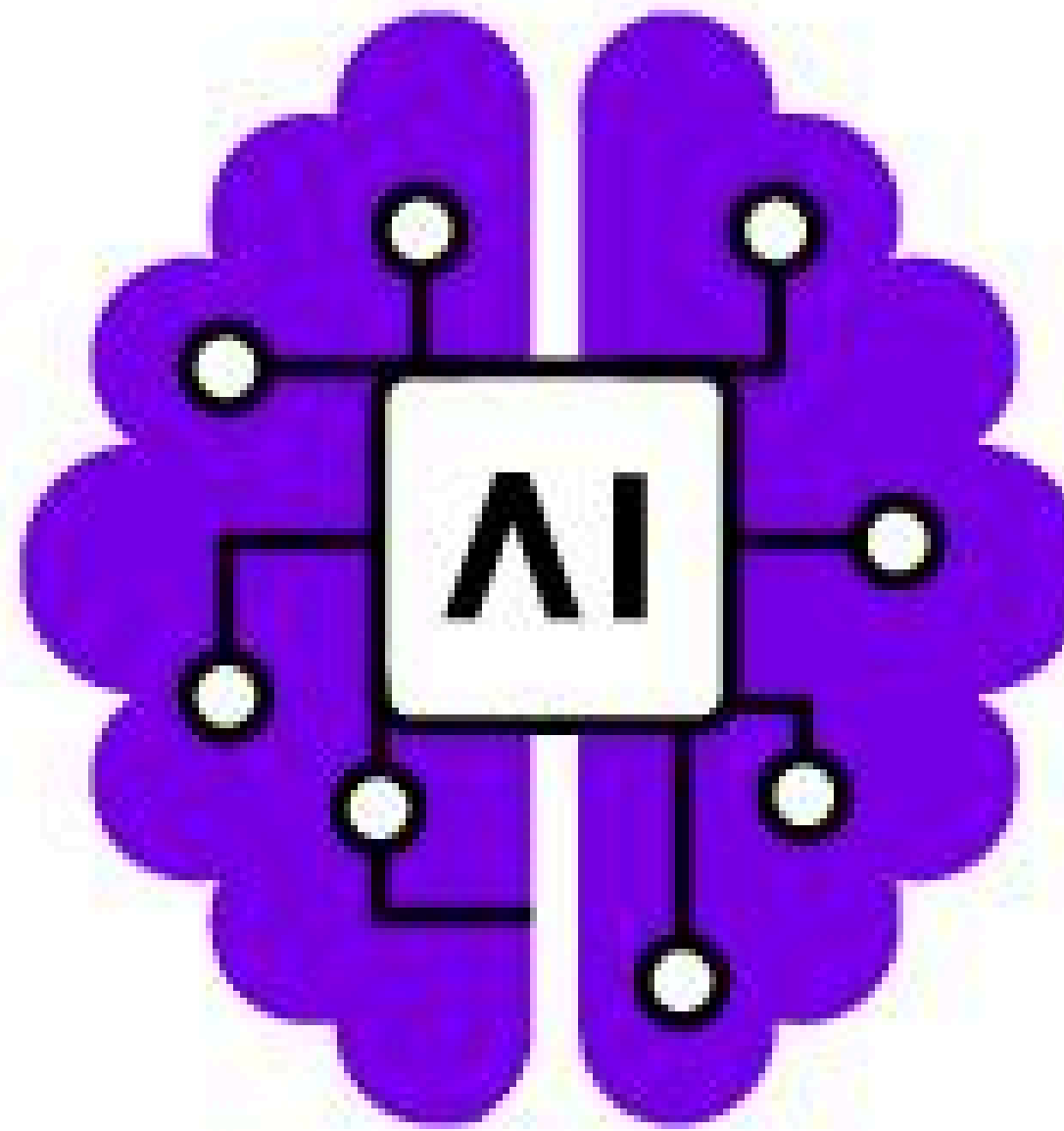
The NIR region extends from 800 to 2500 nm (12 500 to 4000 cm⁻¹; 120 to 375 THz), between the visible from 380 to 780 nm (26 316 to 12 820 cm⁻¹; 385–790 THz) and MIR from 2500 to 15 000 nm (4000 to 400 cm⁻¹; 30 to 120 THz) regions.

Due to the overtone and combination modes and large numbers of possible vibrations, NIR spectra are very complex, consisting of many overlapping peaks (referred to as ‘multicollinearity’), which result in broad bands. This makes it difficult to interpret

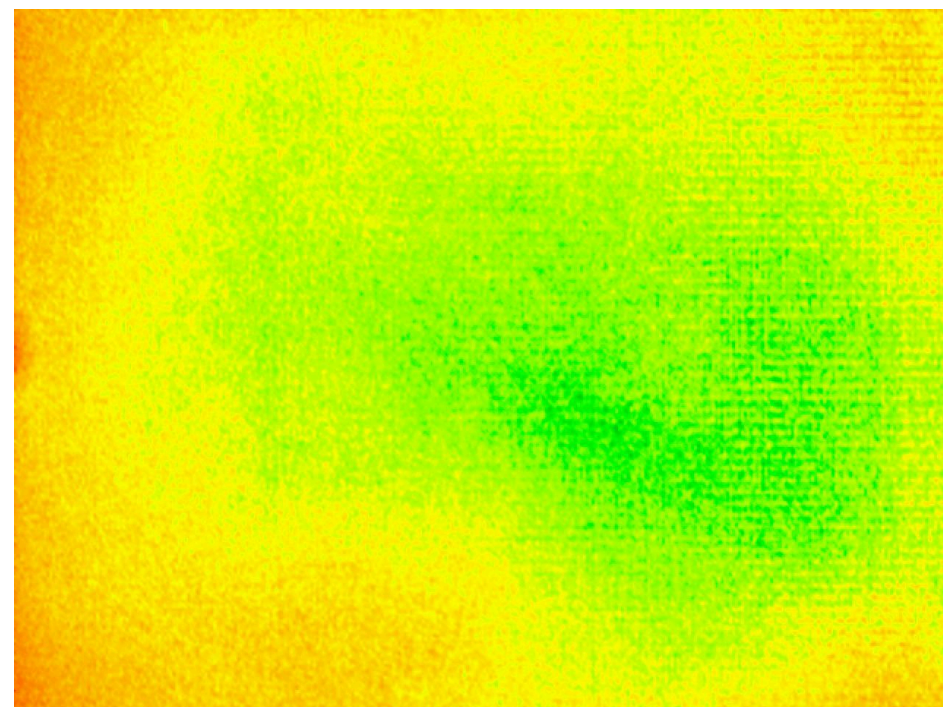
Vibrational Transitions

- Fundamental Band: Transition from the ground vibrational state (v=0) to the first excited state (v=1).
- Overtone Band: Transition from the ground vibrational state (v=0) to a higher excited state (v=2, v=3, etc.).
- Combination Band: Simultaneous excitation of two or more different fundamental vibrations.

Input

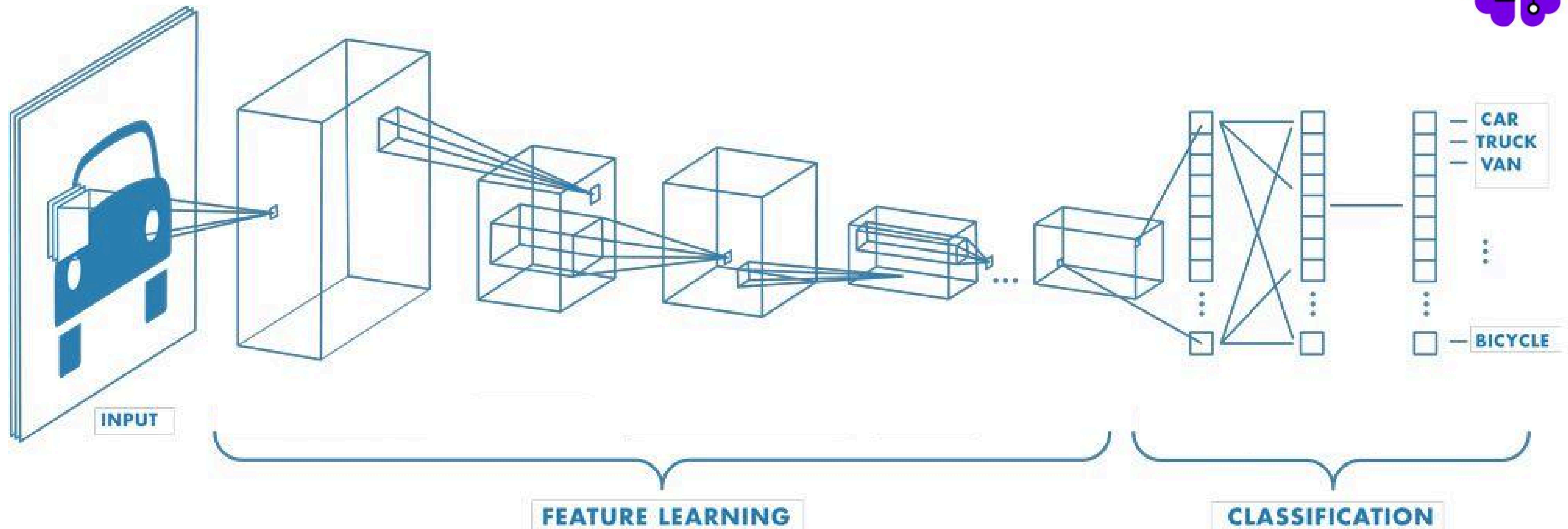
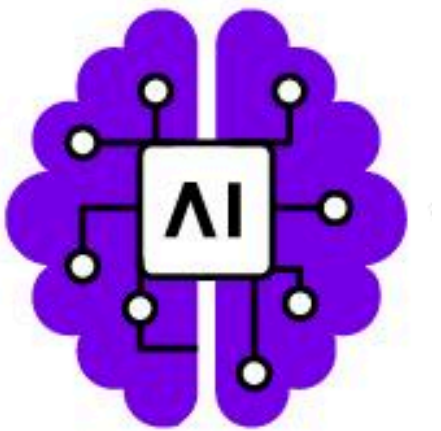


Output



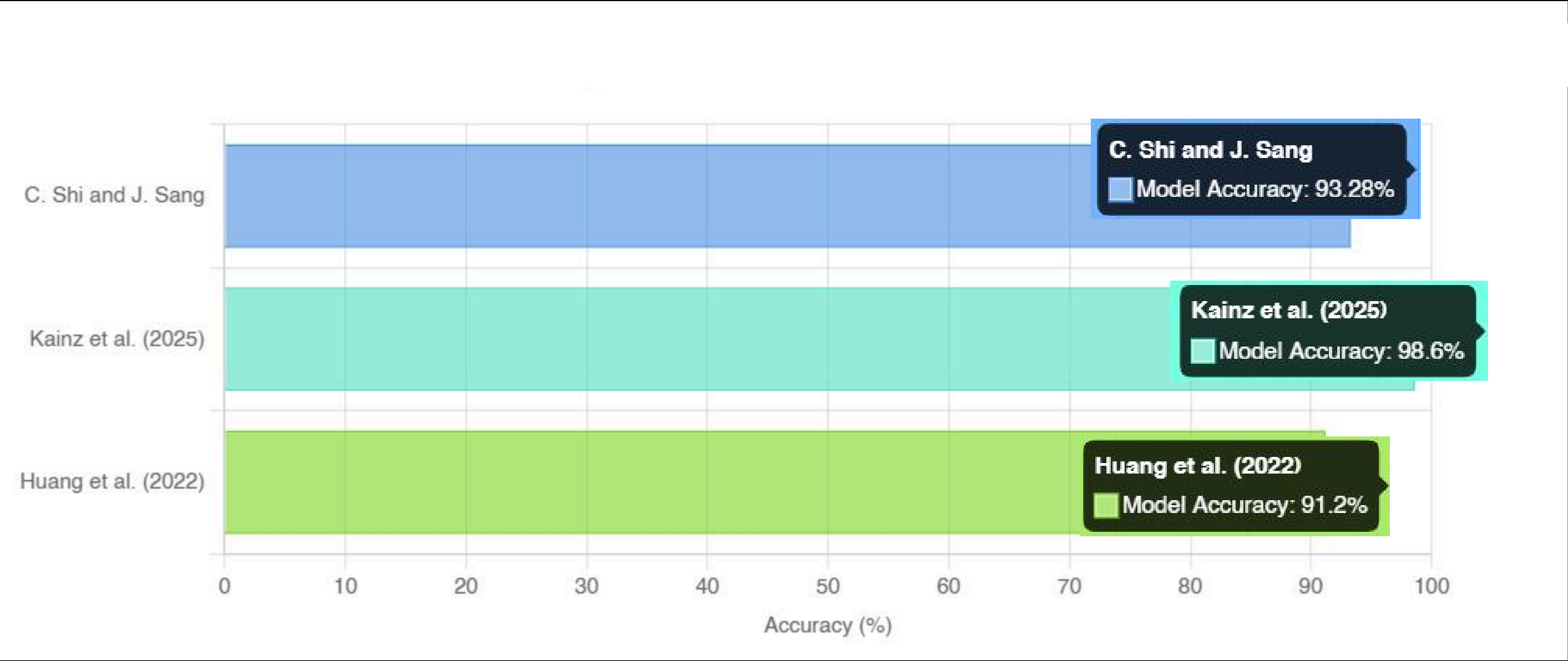
(Cotton)

How Image is Classified using machine learning



Literature Review

Reported Accuracy Levels of Automated Textile Sorting in Literature



Paper	Data Source	Total Spectra Sample	Source Categories
Shi & Sang (2025)	Created their own	11,147	18 (pure & blends)
Huang et al. (2022)	Commercial fibers	600,404	25 (natural & synthetic)
Kainz et al. (2025)	Created their own	Not specified	12 (pure & blends)

Literature Review

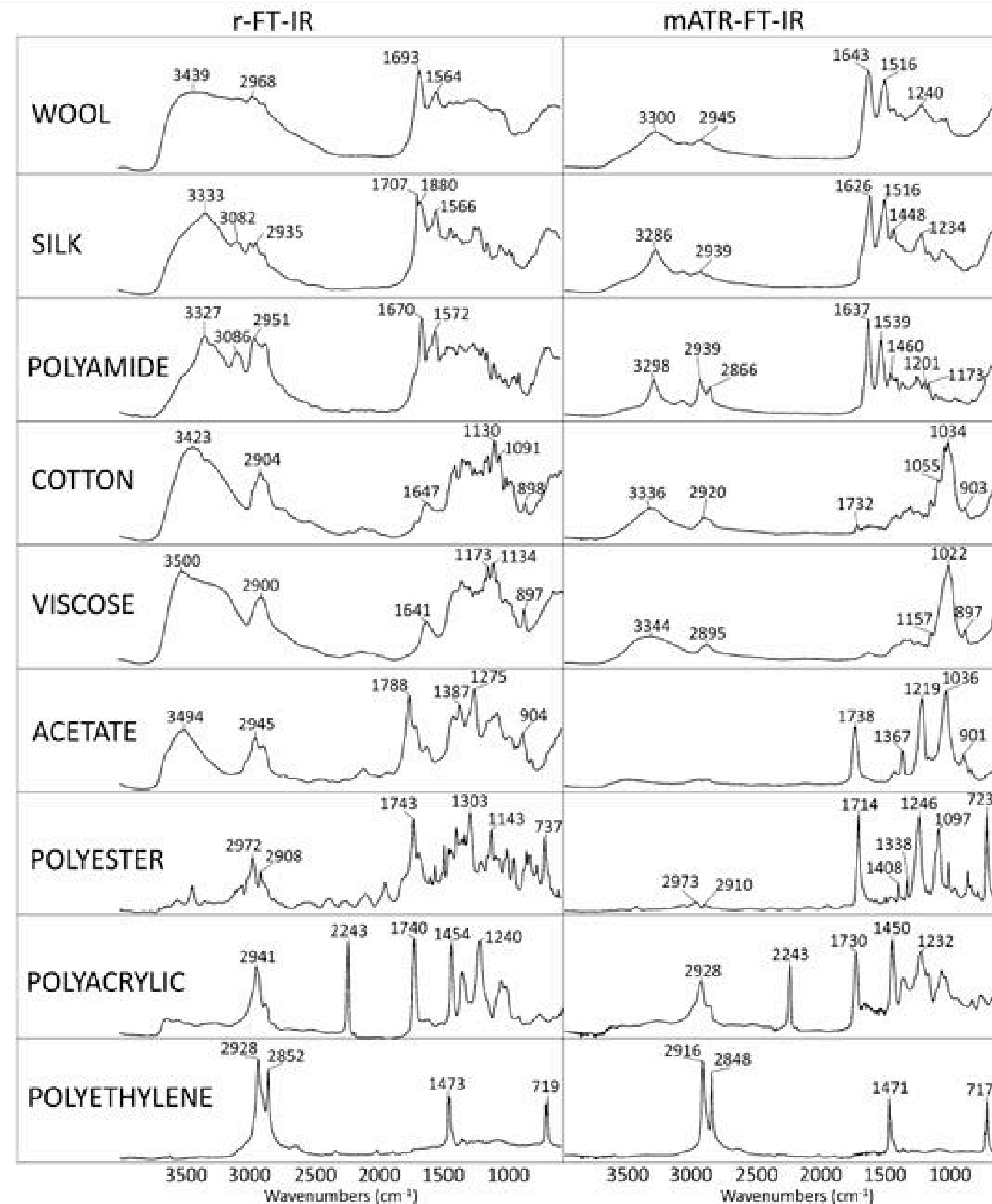


Fig. 1 r-FT-IR and mATR-FT-IR spectra of most common textile fibers. Spectra in figure represent normalized and averaged results of each class. The vertical axis represents absorbance. As natural cellulose-based fibers have very similar spectra, only cotton is presented. Spectra of other natural fibers like linen, jute and sisal are shown in Additional file 1

Fundamental Vibrations by using FT-IR spectroscopy of textile materials

The peaks lie in the range of MIR

Hypothesis

Research Gap

Limited Specificity from Weak NIR Signals:

The inherently weak intensity and broadness of NIR overtone/combination bands significantly limit specific material identification, unlike the strong, specific signals of MIR fundamentals.

Spectral Overlap from Complex Treatments

Pervasive use of multiple dyes, coatings, and finishes introduces significant spectral interferences, leading to ambiguous classification and reduced model robustness.

Research Hypothesis

“Is there any relation between MIR images and the material of textile fabric?”

Objectives

- Verify the material of sample fabric of Cotton, Polyester, Nylon using Flame test and Chemical test
- To create a foundational labelled dataset by collecting Mid-Infrared (MIR) spectral images.
- To train 3 distinct machine learning classifiers : K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and a 1D Convolutional Neural Network
- To systematically evaluate and compare the performance of the three models using key metrics like accuracy, precision, recall, and F1-score
- To determine the most effective model for this classification task.

Methodology (Research Framework)

Fabrics Chosen

3
types

Cotton | Polyester | Nylon

Woven and Knitted both were considered

Dataset Size

120
Images

Each image will be in the file formats .png
or .jpeg

Equipment Required

MIR camera (wavelength 7.5 μ m- 14 μ m)



Experimental Section

Experimental Verification of Fabric Material

Objective: To confirm the identity of unknown fabric samples using physical (burn) and chemical (solubility) tests.

Fibre	Reaction to Flame	Odour	Ash/Residue
Cotton	Burns quickly without melting. Continues to glow after flame is removed.	Burning Paper	Leaves a small, fluffy, grey ash.
Nylon	Fuses, shrinks, and melts. Usually self-extinguishing.	(No distinct odour listed, often celery-like)	Leaves a hard, tough, grey or brown bead.
Polyester	Fuses, shrinks, and melts with black smoke. Usually self-extinguishing.	Chemical	Leaves a hard, tough, round black bead.



Supposed to be nylon but blend



Polyester

Experimental Section

Experimental Verification of Fabric Material Chemical Tests

Cotton

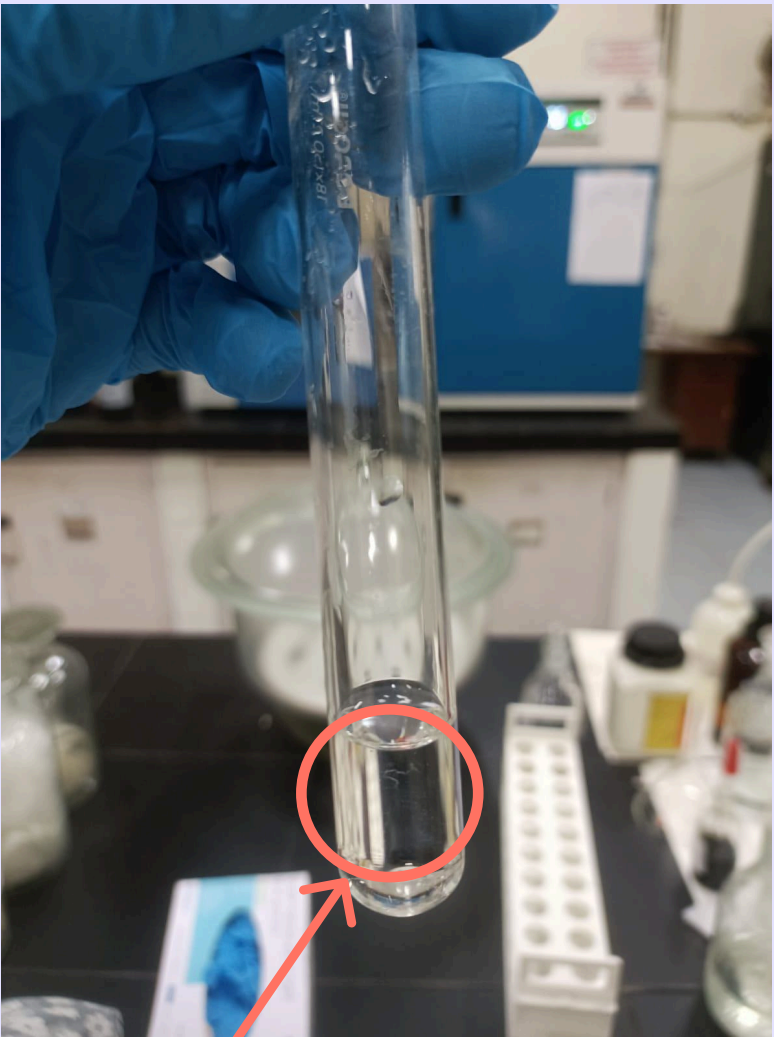
Acetone	Insoluble
5N HCl	Insoluble
59% H ₂ SO ₄	Insoluble
75% H ₂ SO ₄	Soluble

Nylon

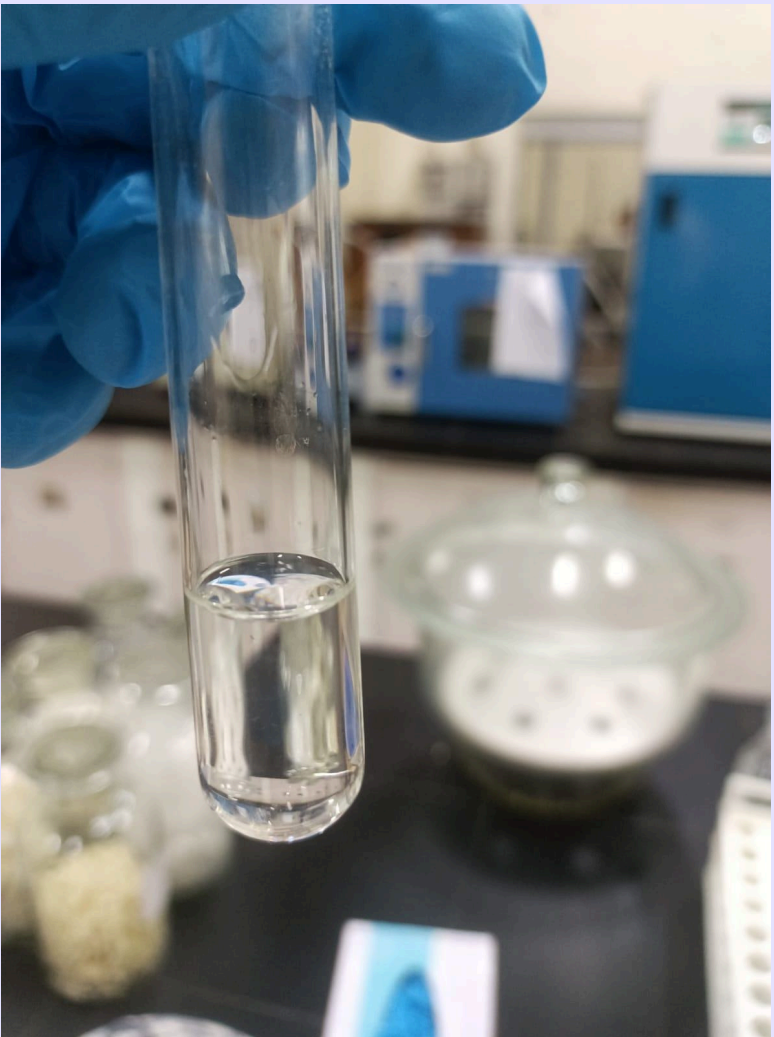
Acetone	Insoluble
5N HCl	Soluble

Polyester

Acetone	Insoluble
5N HCl	Insoluble
59% H ₂ SO ₄	Insoluble
75% H ₂ SO ₄	Insoluble
Conc. H ₂ SO ₄	Soluble



Cotton Sample



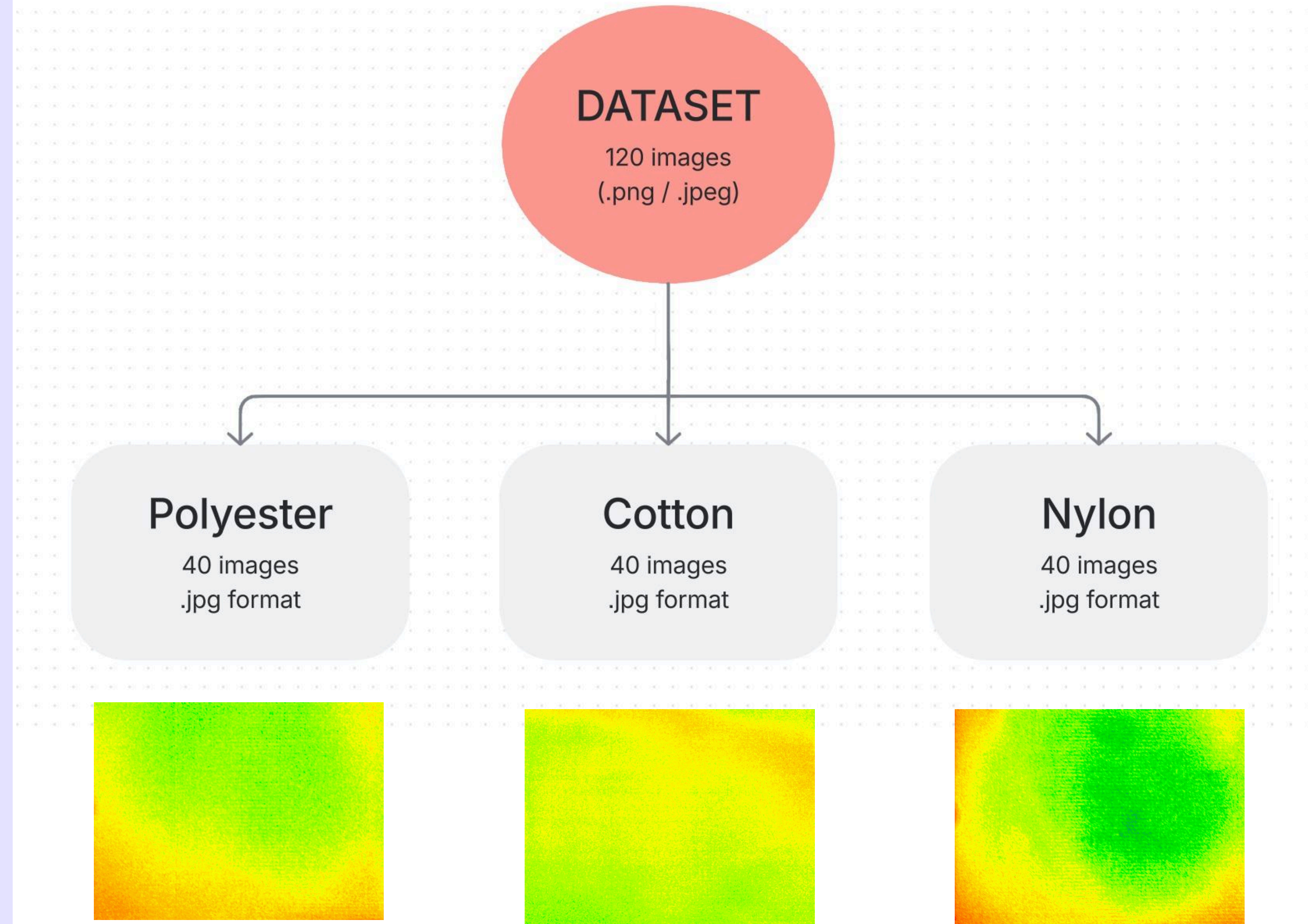
Cotton Sample Dissolved

Current Dataset Overview

MIR Image

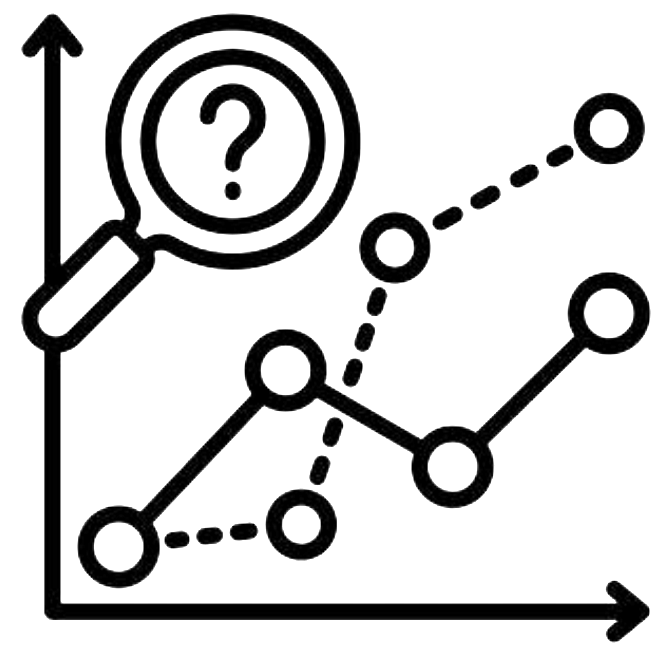
Dimensions : 1280 x 960
Width : 1280 pixels
Height : 960 pixels

- To ensure consistent scale and focus across all samples, a standardized distance of **0.07 meters** was chosen between the camera lens and the sample.
- Throughout the experiment, the environmental temperature was maintained within a controlled range of **21°C to 24°C**.



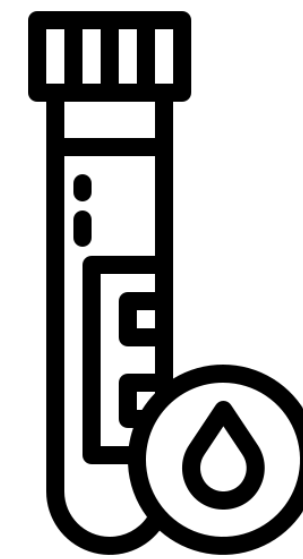
Summary

Gap in Current Research



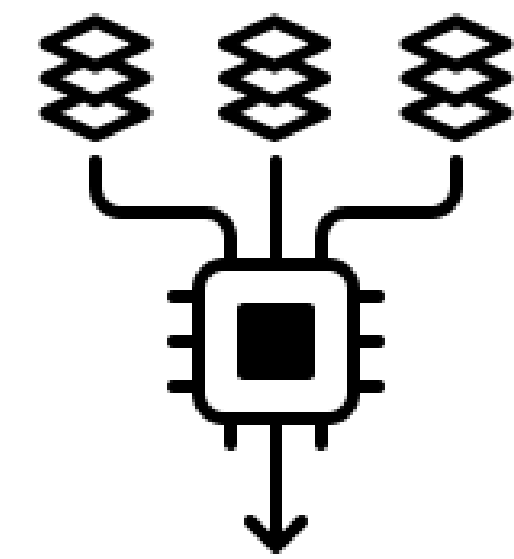
- NIR-based models struggle to distinguish chemically similar fibers.
- Weak NIR Overtones Drive Limited Material Specificity
- Limited exploration of MIR for textiles Fabric.

Progress So Far



- Verified Cotton, Polyester, Nylon using burn tests & chemical solubility tests.
- Created 120 MIR images (40 per fabric type) under controlled conditions.
- Plan to train KNN, SVM, and 1D-CNN on MIR dataset.

Next Steps



Sample Expansion

- Grow dataset with more fabric types, blended textiles, and varied conditions.

Model Training & Testing

- Train and benchmark ML models (KNN, SVM, 1D-CNN).
- Evaluate using accuracy, precision, recall, F1-score.

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